

- 0 3 . 1** Complete **Table 1** by adding a tick (✓) if the feature is present in the biological molecules.

Table 1

Feature	Biological molecule		
	Glycogen	Sucrose	DNA
Contains glucose			
Contains galactose			
Contains fructose			
Contains phosphate			
Formed by condensation reaction(s)			
Is a polymer			

[3 marks]

- 0 3 . 2** Lactase catalyses the hydrolysis of lactose.

A student investigated the effect of galactose on lactase activity.

The student used the following method.

1. Mix together (in tube **X**) solutions of lactose, galactose and lactase in a 15 : 3 : 1 volume ratio so that the total volume is 90 cm³
2. Leave the mixture at 30 °C for 10 minutes.
3. Measure the glucose concentration in the mixture.
4. Repeat steps 1 to 3 with two different mixtures.

Calculate the volume of **galactose** solution used in tube **X**.

[1 mark]

Volume of galactose = _____ cm³

Table 2 shows the student's results.

Table 2

Tube	Mixture	Glucose concentration after 10 minutes / mg dm ⁻³
X	Lactose, galactose, lactase	100
Y	Lactose, water, lactase	180
Z	Galactose, water, lactase	0

0 3 . 3 Galactose is a competitive inhibitor of lactase.

Use this information and your knowledge of enzymes to explain the results obtained in **Table 2**.

[4 marks]

Tube X _____

Tube Y _____

Tube Z _____

Question 3 continues on the next page

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0 3 . 4

Give:

- **one** change the student could make in tube **X** to confirm that galactose is a **competitive** inhibitor
- the result you would expect.

[2 marks]

Change _____

Result _____

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